Aravind Venugopal

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Experience

Blue River Technology, LLC — Jan 2024 - Aug 2024
Machine Learning Engineer, Manager: Devin Matthews — Santa Clara, CA

RBCDSAI, Indian Institute of Technology — Aug 2020 - June 2022
Post Baccalaureate Research Fellow, Advisor: Prof. Balaraman Ravindran — Chennai, India

Robotics Institute, Carnegie Mellon University — Sept 2019 - Feb 2020
Research Intern, Advisor: Prof. Cameron Riviere — Pittsburgh, PA

NEARLab, Politecnico di Milano — June 2019 - Sept 2019
Research Intern, Advisor: Prof. Elena De Momi — Milan, Italy

Education

Ph.D. in Machine Learning — *Aug 2024 - Present*
Carnegie Mellon University, Pittsburgh, PA, USA — CGPA: 4.12/4

M.S. in Machine Learning — *Grad. Dec 2023*
Carnegie Mellon University, Pittsburgh, PA, USA — CGPA: 4.11/4

B.E. (Hons) in C.S. and M.Sc. (Hons) in Economics — *Grad. Aug 2020*
BITS Pilani K. K. Birla Goa Campus, Goa, India — CGPA: 8.53/10 (85.3%)

Merits

- National Talent Search Scholar (*0.5% acceptance out of over 1,000,000*) 2011
- Innovation in Science Pursuit for Inspired Research Scholar (*top 1% nationally in ISC, 12th grade*) 2015

Publications

- **Geometric Time Score Estimators for Offline Reinforcement Learning** [Webpage]
  Aravind Venugopal, Grace Liu, Jeff Schneider, Benjamin Eysenbach
  Under submission.
- **Offline Reinforcement Learning for Plasma Control in Nuclear Fusion: Codebase and Benchmark** [Paper]
  Yang Fu, Haomin Bao, Rohit Sonker, Xiaoyan Hu, **Aravind Venugopal**, Jeff Schneider, Jiayu Chen
  In NeurIPS 2026.
- **Occupancy Reward Shaping: Improving Credit Assignment for Offline Goal-Conditioned Reinforcement Learning** [Paper] [Webpage]
  Aravind Venugopal, Jiayu Chen, Xudong Wu, Chongyi Zheng, Benjamin Eysenbach, Jeff Schneider
  In ICLR 2026.
- **Offline Reinforcement Learning for Rotation and Density Profile Control in Tokamaks** [Paper]
  Rohit Sonker, Hiro Josep Farre Kaga, Jiayu Chen, Andy Rothstein, Ian Char, **Aravind Venugopal**, Namrata Deka, Ricardo Shousha, Egemen Kolemen, Jeff Schneider
  In APS 2026.
- **Policy-Driven World Model Adaptation for Robust Offline Model-based Reinforcement Learning** [Paper]
  Jiayu Chen, Le Xu, **Aravind Venugopal**, Jeff Schneider
  In ICML 2026.
- **MABL: Bi-level Latent-Variable World Model for Sample-Efficient Multi-Agent RL** [Paper]
  Aravind Venugopal, Stephanie Milani, Fei Fang, Balaraman Ravindran

In AAMAS 2024.

- **LaVa: Latent Variable Models for Sample Efficient Multi-Agent Reinforcement Learning** [Paper]
  Aravind Venugopal, Elizabeth Bondi, Fei Fang, Balaraman Ravindran
  In RLDM 2022.
- **Evolutionary Approach to Security Games with Signaling** [Paper]
  Adam Żychowski, Jacek Mańdziuk, Elizabeth Bondi, **Aravind Venugopal**, Milind Tambe, Balaraman Ravindran
  In IJCAI-ECAI 2022.
- **Real-time vessel segmentation and reconstruction for virtual fixtures for an active handheld microneurosurgical instrument** [Paper]
  Aravind Venugopal, Sara Moccia, Simone Foti, Arpita Routray, Robert A. MacLachlan, Alessandro Perin, Leonardo S. Mattos, Alexander K. Yu, Jody Leonardo, Elena De Momi, Cameron N. Riviere
  In International Journal of Computer Assisted Radiology and Surgery, 17(6), 2022.
- **Reinforcement Learning for Unified Allocation and Patrolling in Signaling Games with Uncertainty** [Paper] [Media]
  Aravind Venugopal, Elizabeth Bondi, Harsha Kamarthi, Keval Dholakia, Balaraman Ravindran, Milind Tambe
  In AAMAS 2021.
- **Real-time Compensation of Handheld Surgical Robotic Tool for Safer Neurosurgery**
  Aravind Venugopal, Sara Moccia, Arpita Routray, Simone Foti, Elena De Momi, Cameron N. Riviere
  In IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN), 2020.

Research Projects

Geometric Time Score Estimators for Offline RL

Advisors: Prof. Jeff Schneider and Prof. Benjamin Eysenbach

- We propose novel methods for likelihood evaluation of desired goals under the occupancy measure in RL along probability paths, bridging the gap between representation learning and generative modeling for occupancy density estimation. [Webpage]

Variable Length Action Chunking for Reinforcement Learning

Advisor: Prof. Aviral Kumar

- Developing effective methods for variable-length action chunking with RL for complex long-horizon manipulation tasks.

Occupancy Reward Shaping for Offline Goal-conditioned RL

Advisors: Prof. Jeff Schneider and Prof. Benjamin Eysenbach

- We propose a novel and effective reward-shaping method for credit assignment based on generative modeling of the occupancy measure and optimal transport, demonstrating state-of-the-art performance in offline GCRL. [Paper] [Webpage]

Latent Variable Models for Multi-Agent Reinforcement Learning

Advisors: Prof. Fei Fang, Prof. Balaraman Ravindran

- Developed a latent variable world model for MARL, LaVa, in collaboration with Prof. Elizabeth Bondi, with potential applications to real-world scenarios. [Paper]
- Developed a scalable hierarchical latent-variable model, MABL, that disentangles agent-centric and global information, respectively, from multi-agent trajectories, achieving up to **22x** sample efficiency over the state-of-the-art. [Paper]
- Exploring temporally abstract hierarchical models to use as representation learners for hierarchical RL.

Reinforcement Learning for Unified Allocation and Patrolling in Green Security Games

Advisor: Prof. Balaraman Ravindran

- Developed Combined Security Game Policy Optimization (CombSGPO), combining defender strategies for resource allocation and patrolling in GSGs, collaborating with Prof. Elizabeth Bondi and Prof. Milind Tambe. [Paper]
- Achieved **3x** better utility and **2.5x** better scalability than state-of-the-art RL and MILP-based approaches.
- CombSGPO uses action representation learning, competitive optimization and MARL. It has applications to combat wildlife poaching, illegal logging, illegal fishing etc. [Code]

Real-time Segmentation for Vessel Avoidance with Surgical Robot
Advisors: Prof. Cameron Riviere and Prof. Elena De Momi

- Developed an algorithm integrating real-time segmentation and real-time 3D reconstruction of a blood vessel with a virtual-fixture-based control strategy. [Paper]
- Built segmentation datasets of intra-operative neurosurgery images for training and evaluation.
- Used the control strategy for real-time vessel avoidance by interfacing with Micron, a surgical robot, and conducted real-world trials.
- Achieved vessel avoidance at **33.3 fps** enabling safe operation at up to **400** μ**m** near a vessel. [Code] [Data]

Teaching and Extra-Curriculars

- **Teaching Assistant** for Exec. Education, 17759: Advanced Topics in ML and Game Theory, taught by Prof. Fei Fang, Carnegie Mellon University Aug 2022 - Dec 2022
- **Teaching Assistant** for CS 6700: Reinforcement Learning, taught by Prof. Balaraman Ravindran, Indian Institute of Technology Madras Jan 2022 - May 2022

Academic Services

- **Reviewer:** AAMAS 2024, RLC 2026, ICLR 2027, ACM Health, IEEE Transactions on Signal Processing

Technical Skills

- **Languages and Libraries:** Python, C++, JAX, PyTorch, TensorFlow, ROS, Git
- **Relevant Coursework (Masters and PhD):** Reinforcement Learning (10703), Probabilistic Graphical Models (10708), Computer Vision (16720), Convex Optimization (10725), Probability and Mathematical Statistics (36700), Advanced Introduction to Machine Learning (10715), Machine Learning in Practice (10718)
- **Relevant Coursework (Undergrad):** Probability and Statistics, Econometrics, Machine Learning, Neural Networks and Fuzzy Logic, Logic in Computer Science, Artificial Intelligence, Data Structures and Algorithms, Object Oriented Programming